

Project 1: Threads - Design Document

CSE 421/521 - Operating Systems | Fall 2026

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Alarm Clock is implemented. Priority Scheduling and Advanced Scheduler are planned designs.

Technical references: the supplied course assignment and the Pintos Project 1 manual and 4.4BSD Scheduler appendix.

Pintos Project 1 manual

Pintos 4.4BSD Scheduler appendix

Alarm Clock

A1. Data structures

Copy here the declaration of each new or changed `struct` or `struct` member, global or static variable, `typedef`, or enumeration. Identify the purpose of each in 25 words or less.

Added in devices/timer.c:

```
struct sleep_waiter
{
    uint64_t wake_tick;
    struct semaphore wake;
    struct list_elem elem;
};

static struct list sleepers;
```

sleep_waiter	A sleep request stored on the calling thread's stack.
--------------	---

wake_tick	The tick when the thread can become ready.
-----------	--

wake	A semaphore that blocks the thread until the timer wakes it.
------	--

elem	Connects the request to the sleep list.
------	---

sleepers	The pending requests, sorted by wake_tick.
----------	--

struct thread is unchanged.

A2. Sleep and wakeup

Briefly describe what happens in a call to timer_sleep(), including the effects of the timer interrupt handler.

timer_sleep() returns immediately if the duration is zero or negative. Otherwise, it initializes a local semaphore to zero. With interrupts disabled, it calculates the wakeup tick and inserts the request into the sorted sleep list. It then restores interrupts and calls sema_down() to block.

The timer interrupt checks the front of the list. It removes each request whose wakeup tick has arrived and calls sema_up() on its semaphore. It stops at the first request that is not due. This wakes all threads due on the same tick.

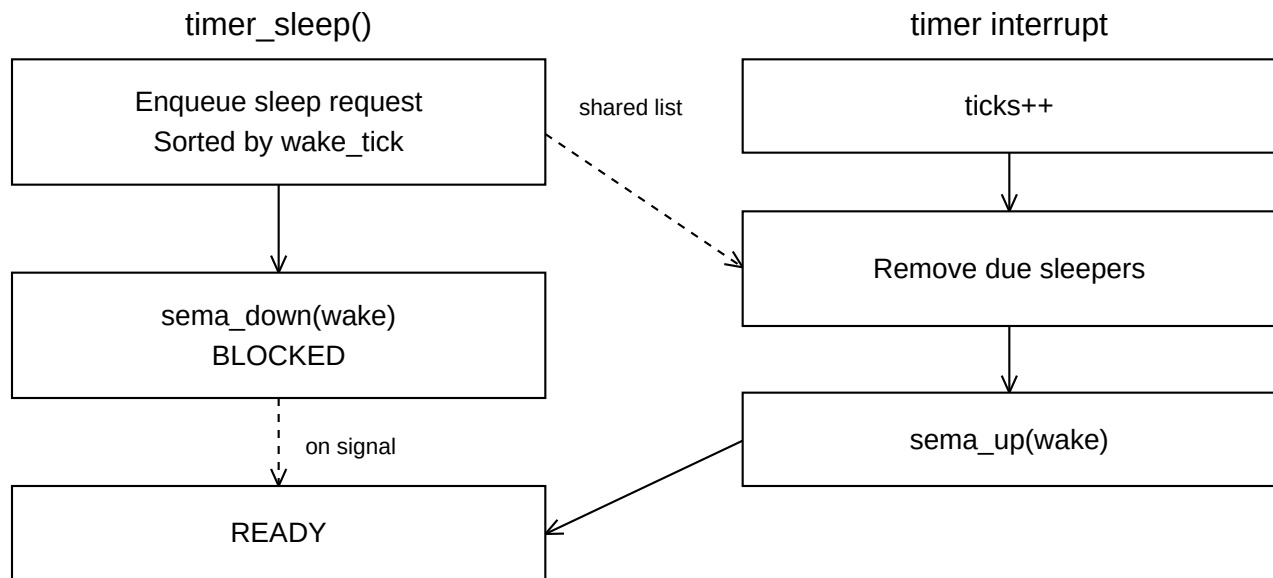


Figure 1. Typical blocking path. A stored early signal lets `sema_down()` return without blocking; a ready thread runs when selected by the scheduler.

A3. Interrupt cost

What steps are taken to minimize the amount of time spent in the timer interrupt handler?

The list is sorted when a thread goes to sleep, so the interrupt handler only visits requests that are due, plus the first one that is not. It does not scan the rest of the list. Inserting a request costs $O(n)$; waking k requests costs $O(k + 1)$.

A4. Concurrent sleep calls

How are race conditions avoided when multiple threads call `timer_sleep()` simultaneously?

Each thread has its own request and semaphore. Interrupts are disabled only while reading ticks and inserting into the shared list. On this single-CPU system, another thread cannot run during those operations. The caller restores interrupts before waiting on its semaphore.

A5. Timer races

How are race conditions avoided when a timer interrupt occurs during a call to `timer_sleep()`?

The request is initialized before it is added to the list. Interrupts are disabled during insertion, so the handler cannot see an unfinished request.

The timer may signal after interrupts are restored but before `sema_down()`. In that case, `sema_up()` leaves the semaphore value at one, and `sema_down()` consumes it without blocking. If `sema_down()` runs first, the thread blocks until the timer signals it.

The handler removes the request before signaling. Its stack storage stays valid while `timer_sleep()` is blocked, and it is no longer in the list when the function returns.

A6. Design choice

Why did you choose this design? In what ways is it superior to another design you considered?

A semaphore lets each thread block and handles a signal that arrives before the thread starts waiting. Keeping the request on the stack avoids allocating memory or adding fields to struct thread.

An unsorted list would make insertion faster, but the timer would have to scan every request on every tick. Sorting at insertion keeps that work out of the interrupt handler. The cost is $O(n)$ insertion with interrupts disabled, which gets more expensive as the list grows.

Priority Scheduling - Proposed Design

B1. Data structures

Copy here the declaration of each new or changed `struct` or `struct` member, global or static variable, `typedef`, or enumeration. Identify the purpose of each in 25 words or less.

Additions to struct thread in threads/thread.h:

```
int base_priority;
struct lock *waiting_lock;
struct list held_locks;
struct list_elem lock_wait_elem;
```

base_priority	The priority requested by the thread before donation.
---------------	---

waiting_lock	The lock the thread is trying to acquire, or NULL.
--------------	--

held_locks	Locks owned by the thread.
------------	----------------------------

lock_wait_elem	Links a pending acquisition into the lock's contenders.
----------------	---

The existing priority field holds the effective priority. thread.h also needs a forward declaration of struct lock.

Additions to struct lock in threads/synch.h:

```
struct list contenders;
struct list_elem held_elem;
```

contenders	Threads whose lock acquisitions have not succeeded yet.
------------	---

held_elem	Links the lock into its holder's held_locks list.
-----------	---

Addition to semaphore_elem in threads/synch.c:

```
struct thread *thread;
```

thread	Identifies the condition waiter, including before it blocks.
--------	--

The ready list and semaphore wait lists keep their existing list links.

B2. Donation tracking

Explain the data structure used to track priority donation. Use ASCII art to diagram a nested donation. (Alternately, submit a .png file.)

held_locks groups donations by the lock being requested. Each lock has a contenders list, and each contender's waiting_lock points to that lock. Together with lock.holder, these links describe the donation chain.

A contender stays on this list until it acquires the lock. This matters because `sema_up()` can make it ready before it has actually taken the lock; it still needs to donate during that interval.

A holder's effective priority is the maximum of its base priority and the effective priorities of contenders for all its held locks. Donation follows `waiting_lock -> holder`, stopping after eight holders, a null holder, or a repeated thread.

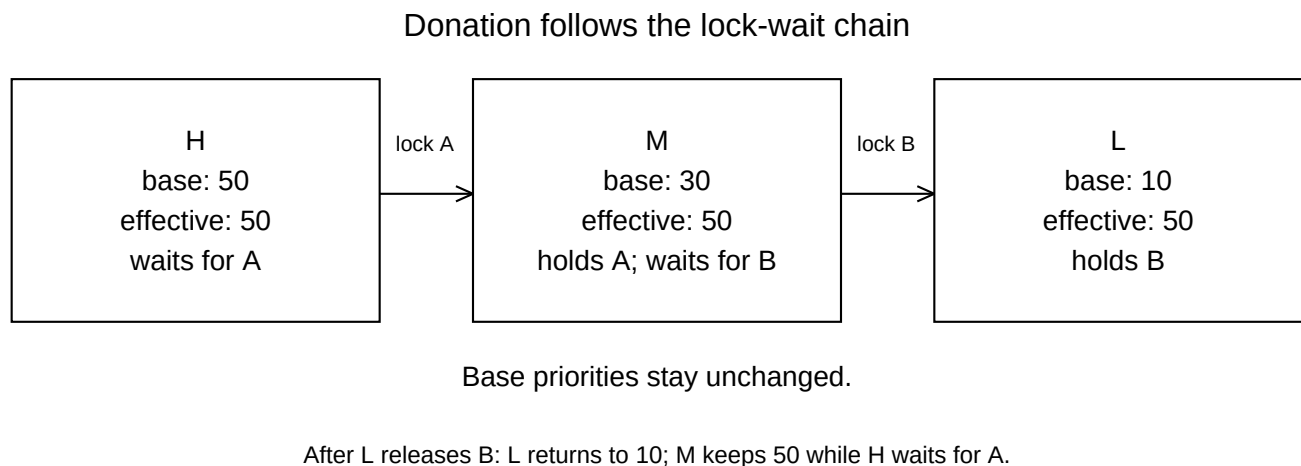


Figure 2. H waits for a lock held by M, and M waits for one held by L. Both holders receive H's priority of 50.

B3. Priority selection and preemption

How do you ensure that the highest priority thread waiting for a lock, semaphore, or condition variable wakes up first?

Ready threads and semaphore waiters are appended at the tail. Selection uses the first maximum of their current effective priorities. This keeps equal-priority threads in FIFO order and handles a donation received while a thread is already waiting.

`sema_up()` selects, removes, and unblocks the highest-priority waiter, then increments the semaphore before allowing preemption. Locks use this same semaphore rule. `cond_signal()` selects through the saved thread pointer. Looking inside its private semaphore is unsafe because the signal can arrive before `sema_down()`. An early signal stays in the semaphore; a blocked waiter becomes ready. `cond_wait()` reacquires the monitor lock on return.

A higher-priority ready thread triggers a yield after the caller finishes its state changes. `thread_create()` performs the same check for a new thread. Idle gives way to any ready thread, including priority zero. `thread_unblock()` keeps its atomic contract in thread context, so its caller checks preemption after finishing. Interrupt handlers request `intr_yield_on_return()` instead of switching inside the handler.

B4. Lock acquisition

Describe the sequence of events when a call to `lock_acquire()` causes a priority donation. How is nested donation handled?

If the lock is busy, `lock_acquire()` sets `waiting_lock`, registers the caller in contenders, and recomputes the holder's effective priority. If that holder is waiting for another lock, the update continues along the holder chain, up to eight holders. `sema_down()` then blocks the caller until it can take the semaphore.

After success, the caller removes its contender entry if it registered one, clears `waiting_lock`, and becomes the holder. The lock is added to `held_locks`, and the new holder's priority is recomputed from remaining contenders. Ownership bookkeeping also applies to an uncontended acquire and a successful `lock_try_acquire()`. A failed try

registers no wait or donation. The list and priority changes occur within the synchronization primitives' short atomic sections.

B5. Lock release

Describe the sequence of events when `lock_release()` is called on a lock that a higher-priority thread is waiting for.

The releasing thread removes the lock from `held_locks` and clears its holder. It recomputes its effective priority from `base_priority` and the contenders for its remaining locks. Donations for the released lock are removed; donations for other locks still count.

`sema_up()` then unblocks the highest-priority waiter and releases the semaphore. The next holder recomputes its priority from the remaining contenders when acquisition succeeds. Once release bookkeeping is complete, the caller yields if a higher-priority thread is ready.

B6. Priority-update race

Describe a potential race in `thread_set_priority()` and explain how your implementation avoids it. Can you use a lock to avoid this race?

Suppose `thread_set_priority()` reads the donation list and is preempted before writing its result. Another thread adds a higher donation. When the setter resumes, its stale result could overwrite the new priority.

The base-priority change, donation scan, effective-priority write, and ready-priority comparison belong in one short atomic update. An active donation can keep the effective priority above the new base value. Any yield follows the completed update; `thread_get_priority()` returns the effective value.

A sleeping lock would create another scheduling and donation dependency inside the mechanism being updated, and an interrupt handler cannot acquire it. Short interrupt exclusion protects this internal state on Pintos's single CPU. Actual waits still use semaphores and locks.

B7. Design choice

Why did you choose this design? In what ways is it superior to another design you considered?

Tracking contenders by lock makes it easy to remove just the donations associated with a released lock. Selecting by current priority also avoids moving a thread whenever a donation changes its position in a ready or waiting queue.

An ordered queue would make selection cheaper, but every priority change would require fixing the thread's position, even when it is blocked on another semaphore. Our list-based design has fewer update paths. Its cost is linear selection, scans of held locks and contenders, and a few extra list links per thread and lock.

Advanced Scheduler - Proposed Design

C1. Data structures

Copy here the declaration of each new or changed `struct` or `enum` member, global or static variable, `typedef`, or enumeration. Identify the purpose of each in 25 words or less.

In `threads/fixed-point.h`:

```
typedef int32_t fixed_t;
#define FP_SHIFT 14
#define FP_SCALE (1 << FP_SHIFT)
```

<code>fixed_t</code>	A signed 17.14 fixed-point value.
----------------------	-----------------------------------

FP_SHIFT	The number of fractional bits.
FP_SCALE	The scale factor for integer and fixed-point conversions.

Additions to struct thread in threads/thread.h:

```
int nice;
fixed_t recent_cpu;
```

nice	The thread's niceness value, inherited from its creator.
recent_cpu	Recent CPU usage, including the fractional part.

In threads/thread.c:

```
static fixed_t load_avg;
```

load_avg	The moving average of runnable non-idle threads.
----------	--

priority, ready_list, all_list, and thread_mlfqs are reused. Selection reads current priorities from the single ready list. Fixed-point multiplication and division use int64_t intermediate values.

The initial thread starts with nice=0 and recent_cpu=0. New ordinary threads inherit both values before being added to all_list. Idle is identified before it becomes ready and excluded from CPU and load accounting. load_avg starts at zero, and idle keeps PRI_MIN priority. MLFQS ignores manually supplied priorities and bypasses donation. Changing nice recalculates priority and checks preemption.

C2. Scheduling example

Suppose threads A, B, and C have nice values 0, 1, and 2. Each has a recent_cpu value of 0. Fill in the table below showing the scheduling decision and the priority and recent_cpu values for each thread after each given number of timer ticks:

Tick	recent_cpu			Priority			Next thread
	A	B	C	A	B	C	
0	0	0	0	63	61	59	A
4	4	0	0	62	61	59	A
8	8	0	0	61	61	59	B
12	8	4	0	61	60	59	A
16	12	4	0	60	60	59	B
20	12	8	0	60	59	59	A
24	16	8	0	59	59	59	C
28	16	8	4	59	59	58	B
32	16	12	4	59	58	58	A
36	20	12	4	58	58	58	C

C3. Table conventions

Did any ambiguities in the scheduler specification make values in the table uncertain? If so, what rule did you use to resolve them? Does this match the behavior of your scheduler?

A, B, and C start ready in that order and remain runnable. Each row shows values after the listed tick's accounting and priority update. The final column gives the thread selected for the next four-tick slice.

At slice expiry, the running thread joins the ready-list tail. The first maximum priority wins. Thus B precedes A at tick 8, and C precedes both at tick 24. A is selected again at tick 4 because its priority is still higher. These conventions are also the rules for the planned scheduler. With `TIMER_FREQ=100`, no one-second update occurs in this table.

C4. Update order and performance

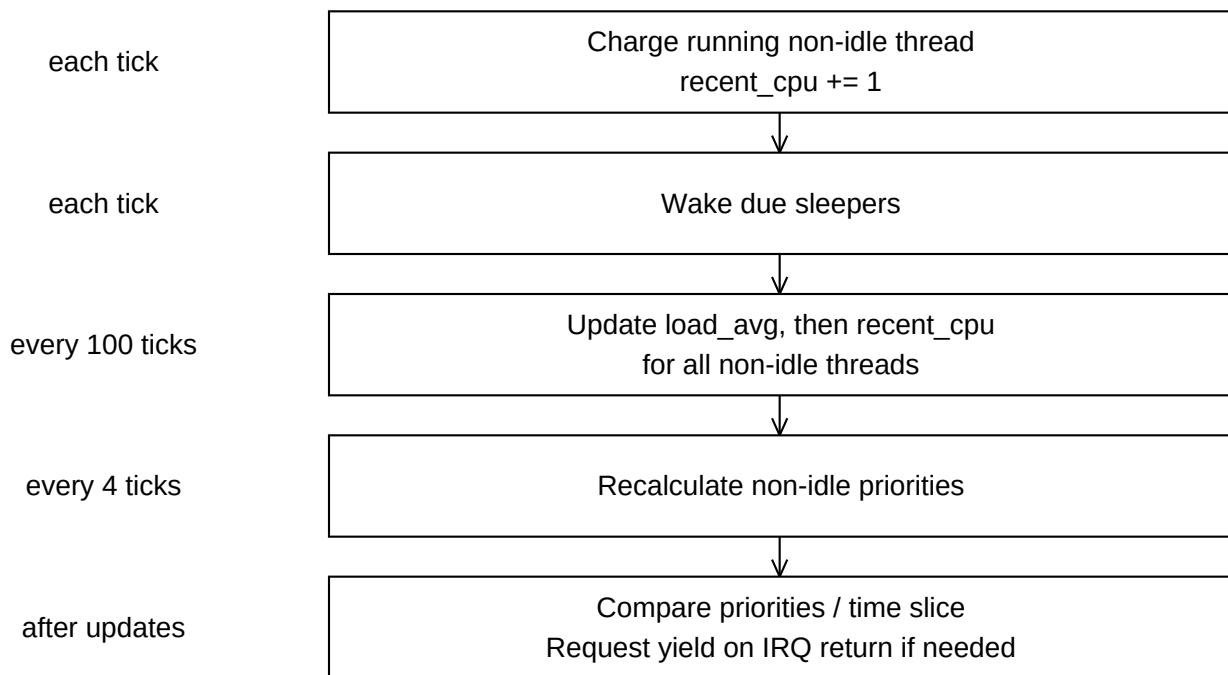
How is the way you divided the cost of scheduling between code inside and outside interrupt context likely to affect performance?

Every tick adds one `recent_cpu` unit to the running non-idle thread, then wakes due sleepers. Every `TIMER_FREQ` ticks, `load_avg` is updated before `recent_cpu` is recomputed for all non-idle threads, including blocked threads. Every fourth global tick updates their priorities. When both boundaries coincide, the priority formula uses the new values.

```
priority = PRI_MAX - recent_cpu / 4 - 2 * nice
load_avg = (59/60) * load_avg + (1/60) * ready_threads
recent_cpu = (2*load_avg)/(2*load_avg+1) * recent_cpu + nice
```

`ready_threads` counts the ready list plus the running non-idle thread, after alarm wakeups. The updates finish before interrupt return. This requires splitting `thread_tick()` into accounting and periodic updates, with the latter called after the alarm wake loop. After the updates, the handler compares the current and ready priorities and requests `intr_yield_on_return()` when needed.

Periodic updates cost $O(n)$ for n threads. Dispatch costs $O(r)$ for r ready threads, and waking k sleepers adds $O(k+1)$. All-thread scans occur only at the specified boundaries. Thread creation and `thread_set_nice()` calculate one thread's priority outside interrupt context, with short atomic sections for shared state.



Use the intervals on the left; skipped steps continue downward.

Figure 3. Timer-update order with the default `TIMER_FREQ` of 100. Only non-idle threads participate in CPU and load accounting.

C5. Design tradeoffs

Briefly critique your design, pointing out advantages and disadvantages in your design choices. If you were to have extra time to work on this part of the project, how might you choose to refine or improve your design?

One ready list keeps selection simple and works with both schedulers. Reading priorities at dispatch avoids an ordering that becomes stale after a periodic update. Blocked threads receive updated values without being moved between queues.

The main costs are $O(r)$ dispatch and $O(n)$ periodic scans. With more time, we would consider 64 FIFO ready queues to reduce dispatch cost. That would require moving ready threads after priority changes and preserving the same tie order. For this project, the single list is the simpler starting point.

C6. Fixed-point arithmetic

The assignment explains arithmetic for fixed-point math in detail, but it leaves it open to you to implement it. Why did you decide to implement it the way you did? If you created an abstraction layer for fixed-point math, that is, an abstract data type and/or a set of functions or macros to manipulate fixed-point numbers, why did you do so? If not, why not?

We plan to put small inline helpers in `threads/fixed-point.h` using signed 17.14 values and $F = 1 \ll 14$. Naming the operations keeps the scale conversions in one place and makes the scheduler formulas easier to check.

An integer n becomes $n * F$, and adding it to x uses $x + n * F$. Two fixed-point values multiply as $((\text{int64_t } x * y) / F$ and divide as $((\text{int64_t } x * F) / y$. The wide intermediate prevents scaling overflow. Multiplication and division by F also handle negative operands without relying on signed shifts.

Stored values keep their fractions, including negative `recent_cpu`. Priority is calculated in fixed point, then converted toward zero and clamped to 0..63. For a valid positive priority, this rounds down. The `recent_cpu/4` term is not truncated early. In the one-second update, the coefficient is calculated before multiplying it by `recent_cpu`.

The load and recent CPU getters multiply by 100 in a wide intermediate and round to nearest. For scaled x , use $(x + F/2)/F$ when x is nonnegative and $(x - F/2)/F$ when negative, with division toward zero. Separate helpers for truncation and rounding make that distinction explicit.

Optional survey questions

In your opinion, was this assignment, or any one of the three problems in it, too easy or too hard? Did it take too long or too little time?

Did you find that working on a particular part of the assignment gave you greater insight into some aspect of OS design?

Is there some particular fact or hint we should give students in future quarters to help them solve the problems? Conversely, did you find any of our guidance to be misleading?

Do you have any suggestions for the TAs to more effectively assist students, either for future quarters or the remaining projects?

Any other comments?